

MLOps

Homework 1

Due Friday, September 25th, at 11:59:59 pm

1. git
 - a. Create a git repository. If you have one already, create a new repository for mlops.
 - b. Take the sampregdata and build a linear regression model with scikit-learn, using the 1 X that fits the best. Put the model and the data in a proper organization.
 - c. Create a new model with 2 X's. Use proper version control to designate this model as the current model while still have the previous model as accessible.
 - d. Grant access to one person in the class. That person should be able to access the data and run the models.
 - e. Create a readme file which documents the most recent model and compares it to the previous one. 1 paragraph is sufficient for this.
2. The first set of slides mention demand forecasting. For this, suppose you are working with “demand planners” to create demand forecasts for a set of 100 products. They products are small, tangible goods.
 - a. What is a demand planner?
 - b. Why would a demand planner need product forecasts?
 - c. Based on searching, what is the time frequency most likely needed for these forecasts?
 - d. What data do you need to get started?
 - e. You are to meet with a couple of the demand planners. What are some questions you would like to ask them?
 - f. After building the models, how would you show to the demand planners that the results should be trusted?
 - g. Describe some characteristics of the data that would be needed to productionize the models.
 - h. What is your best guess as to how they would want to receive the model results?
 - i. What sources did you use to find these answers?
3. You are in charge of making a holiday dinner, using one oven and four burners. Two cooks must prepare the following eight items. Both cooks can work at the same time, but a task requiring continuous attention occupies one cook for its entire active time.
 - a. **Gravy:** 12 minutes of continuous work. It cannot begin until the roast is finished because it uses the pan drippings. It requires one burner.
 - b. **Roast beef:** 15 minutes of active preparation, followed by 70 minutes unattended in the oven. After leaving the oven, it must rest for at least 15 minutes before serving.

- c. **Mashed potatoes:** 10 minutes of active preparation, followed by 25 minutes unattended boiling on one burner, followed by 8 minutes of active mashing and seasoning.
- d. **Dinner rolls:** 8 minutes of active preparation, followed by 15 minutes unattended in the oven.
- e. **Green beans:** 18 minutes of continuous preparation and cooking using one burner.
- f. **Baked macaroni and cheese:** 20 minutes of active preparation, including 8 minutes on one burner, followed by 30 minutes unattended in the oven.
- g. **Glazed carrots:** 8 minutes of active preparation, followed by 20 minutes unattended cooking on one burner, followed by 5 minutes of active finishing.
- Salad:** 17 minutes of continuous preparation and does not require the oven or stove.

The oven can hold the roast and only one additional item at the same time. There are four burners. Unattended boiling or simmering does not occupy a cook. **Determine a schedule** for the two cooks that minimizes the time until the entire meal is ready to serve. Indicate when each task begins and ends, which unattended cooking periods overlap with other work, and the earliest possible serving time, **using a Gantt chart**, color-coded by cook.

4. A bank wants to build a model that flags potentially fraudulent credit-card transactions. Transactions occur in real time, but fraud labels may arrive days or weeks later through customer reports or investigator decisions. Investigators tend to review transactions that are already considered suspicious.
 - a. Draw the model-development process, beginning with historical transaction data. Show how transactions are selected, labels are created, data are prepared, and training and evaluation data are formed. Identify problems with target leakage.
 - b. Show how candidate models are developed, evaluated, and compared. Indicate what must happen before a model is approved for deployment.
 - c. Extend the diagram into deployment. Show how a new transaction enters the system, receives a fraud score, and may be flagged for investigation. Be as specific as possible for the data flows.
 - d. Show which information is available when the model makes its prediction, and which information only becomes available later.
 - e. Show how customer reports and investigator decisions eventually produce new fraud labels. Some transactions are investigated while others are not; indicate how this could affect the data collected.
5. Process time simulation
 - a. Take the demand forecasting process diagram from the slides. We will simulate with the following random variables for each step
 - i. SH Analysis: Exponential, mean of 10 days
 - ii. Data Extract: Normal, mean 15 days, standard deviation 3 days
 - iii. Dev Exploration: Normal, mean 3 days, standard deviation $\frac{1}{2}$ day
 - iv. Dev Modeling: Poisson distribution, mean 30 days

- v. Process Integration: Normal, mean 50 days, standard deviation 20 days
 - vi. Data Pipeline: Poisson distribution, mean 100 days
 - vii. Model Productionization: Normal distribution, mean 30 days, standard deviation 5 days
- b. Perform 1,000,000 simulations of this process, taking care to account for the dependencies.
- 1. What is the average and median amount of time for the entire process? What is a 95% (2-sided) confidence interval?
 - 2. Which step contributes the most towards the overall time and the uncertainty in the overall time?
- ii. Suppose the data engineer who is building the data pipeline leaves the company and the mean value becomes 150 days to build it. What is the average and 95% confidence interval for the length of the delay?